

Enhancing ML-based anomaly detection in data management for security through integration of IoT, cloud, and edge computing

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ABSTRACT

The widespread adoption of cloud computing, edge computing, and the Internet of Things across various domains has enhanced data management and automation capabilities. However, large-scale IoT implementations face considerable challenges, including concerns about data quality, security risks, and the need to identify anomalies. This study introduces a multi-tiered machine learning-based approach to detect anomalies, specifically targeting security threats, performance irregularities, and sensor malfunctions within IoT-Edge-Cloud ecosystems. The proposed system improves detection precision, response times, and overall security resilience by incorporating XAI for transparent decision-making and a layered security approach to mitigate threats. The framework operates on several levels: (i) the IoT Layer, where secure microcontrollers collect and transmit sensor data; (ii) the Edge/Fog Layer, which conducts pre-processing and real-time filtering to minimize cloud reliance; (iii) the Cloud Layer, where ML-based anomaly detection algorithms, such as the Isolation Forest and Local Outlier Factor, examine data; and (iv) the Smart Single-Page Application architecture that integrates IoT-Edge-Cloud ecosystem, which offers low-latency visualization, secure data transmission, and interactive anomaly monitoring. Furthermore, XAI techniques improve interpretability by identifying key factors that influence anomaly classification and increase transparency for security analysts. A case study in IoT-Healthcare settings validated the efficacy of the proposed system in identifying network intrusions, sensor failures, and operational anomalies, achieving an AUROC score of 1.00 using an isolated forest. Comparative assessments demonstrate that this approach surpasses existing anomaly detection solutions by enhancing detection accuracy, decreasing latency through edge processing, and improving explainability with AI integration. The study concludes by exploring the challenges and advantages of combining IoT, cloud and edge computing for secure and scalable anomaly detection, thus providing insight into optimal database management and security strategies for IoT-cloud interactions.

1. Introduction

The Internet of Things (IoT), Cloud Computing (CC), Edge Computing (EC), and Fog Computing (FC) have transformed data management by enabling real-time monitoring, automation, and analytics. However, the widespread adoption of the IoT has introduced significant hurdles to ensure data security, maintain quality, and detect anomalies. IoT devices generate vast amounts of diverse data, often requiring immediate processing to identify security threats, sensor malfunctions, and operational anomalies.

Although CC provides scalable and flexible computational resources, it raises concerns about data security and transparency. Simply storing IoT data in cloud-based SQL or NoSQL databases is insufficient; carefully designed, robust, and secure connections between IoT devices, cloud infrastructure, and edge systems are essential to protect against unauthorized access and potential data breaches. Although edge and fog

computing can help address latency issues by processing information closer to its source, the security implications of these approaches remain understudied.

This study focuses on the protocols and security measures that govern microcontrollers responsible for generating and transmitting data to the cloud. In addition, it aims to enable real-time data retrieval from local server storage and to enhance anomaly detection through meticulous analysis. The approach involves leveraging Machine Learning (ML) and Explainable AI (XAI) to enhance anomaly detection capabilities in data management by seamlessly integrating the IoT and CC. Furthermore, the study seeks to investigate the advantages and challenges associated with the integration of IoT and cloud technologies. The research evaluated security measures in database management for IoT devices and CC and EC systems, providing valuable information on optimal methodologies for implementing database management and security measures in the context of the interaction between IoT and cloud and

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edge technology. Through rigorous evaluation and experimentation, we substantiate the practical applicability and viability of our proposed solution.

In today's data-driven society, effective data management is essential for ensuring the integrity, availability, and confidentiality of digital information. In recent years, rapid advances in technology, particularly proliferation of IoT, CC, and EC, have transformed industries and created new opportunities in various domains (Nasser et al., 2021). The increasing affordability and intelligence-enhancing capabilities of IoT devices have accelerated their adoption. These devices, equipped with advanced sensor technology, now form vast interconnected networks that enable real-time communication and automation. Similarly, CC has emerged as a critical platform for storing, processing, and distributing data on a large scale (Puliafito et al., 2015). However, the integration of IoT, CC, and EC presents unique challenges, particularly in data management and security, which constitute the core focus of this study (Boukerche & Coutinho, 2021).

The integration of IoT and CC for data exchange, storage, and processing introduces several challenges. This combination often results in longer data processing times, delays in data transmission, and a rapid increase in data production. Although IoT-Cloud integration provides scalability and computational benefits, it requires a reliable network connection and effective database management approaches (Yazici & Taşkomaz, 2024), especially for large-scale database applications. Moreover, this convergence creates potential security risks, such as information leaks, cyber threats, and Distributed Denial of Service (DDoS) attacks (Boukerche & Coutinho, 2021). These concerns underscore the urgent need for thorough research into data administration, security weaknesses, and anomaly identification within IoT-Cloud interactions. Table 1 provides a classification of the main anomaly types that may occur in IoT-Edge-Cloud-based systems.

While IoT-edge-cloud architectures show promise, secure and efficient anomaly detection continues to pose a significant research challenge. Current methods face several issues:

- Security risks in IoT cloud communication - IoT data transmission often lacks adequate security measures, exposing systems to threats such as man-in-the-middle attacks, unauthorized access, and data breaches.
- Delayed anomaly detection - Cloud-based approaches introduce latency issues, rendering real-time anomaly detection unfeasible for time-sensitive applications.
- Opaque ML-based anomaly detection - Conventional black-box ML models achieve high detection accuracy but lack transparency, making it difficult for system administrators to comprehend the reasons behind flagged anomalies.
- Under-utilization of Edge/Fog Computing for enhanced security - Although Edge/Fog Computing reduces latency, its potential security benefits and role in anomaly detection have not been fully explored.

To tackle the challenges outlined, this study introduces a multilayered ML-based anomaly detection system built on IoT, Edge, Fog, and cloud computing to deliver real-time, secure, and interpretable anomaly detection. The system consists of four key layers: (i) a secure real-time data collection in the IoT layer, (ii) a pre-processing and local anomaly detection in the edge / fog layer, (iii) a scalable ML-based anomaly classification in the cloud layer, and (iv) a Smart Single-Page Application (SSPA) for security monitoring and anomaly visualization.

By leveraging EC, the system reduces latency in anomaly detection while improving security through multi-layered encryption and authentication processes. Furthermore, using XAI via Shapley Additive Explanations (SHAP), security analysts gain insights into the reasoning behind the detection of specific anomalies, which enhances decision-making and transparency within the system. This hybrid approach achieves low-latency, high-accuracy, and interpretable anomaly detection, making it highly relevant to applicable to real-world IoT security scenarios.

The key objectives of this research are as follows:

- Develop a multi-layered ML-based anomaly detection framework that ensures efficient, secure, and real-time anomaly detection in IoT-Edge-Cloud environments, leveraging the SSPA for rapid data processing and secure transmission.
- Enhance security and database management by assessing and optimizing current protocols for IoT and cloud data handling, providing insights into robust security measures for interconnected systems.
- Integrate XAI techniques to improve interpretability and transparency in ML-based anomaly detection, offering actionable insights into IoT system behavior and decision-making.
- Validate the framework in real-world IoT environments, demonstrating its effectiveness, scalability, and robustness in industrial settings through experimental analysis.

This research aims to enable organizations to confidently implement IoT-CC technologies while addressing security concerns and ensuring effective and scalable anomaly detection. The paper is structured as follows. Section 2 offers a detailed examination of the existing literature, exploring current approaches to anomaly detection, IoT security issues, and ML-based methods. Section 3 outlines the research approach, encompassing data collection, ML implementation, and integration with IoT-CC systems. Section 4 presents and evaluates the experimental results, highlighting the efficacy of the proposed strategy. Section 5 explores the challenges and constraints of the study. Lastly, Section 6 summarizes the research by summarizing key findings and discussing their impact on data management and security.

2. Review of the literature

This section presents a comprehensive review of existing studies on the integration of IoT, CC, EC, and ML-based anomaly detection. The objective is to highlight the current advances, challenges and research gaps that motivate this study.

2.1. Literature review methodology

This study employs a systematic approach to its literature review, ensuring thorough coverage of pertinent research on IoT-edge-cloud security, ML-based anomaly detection and XAI. The selection process involved a structured search using specific keywords in respected academic databases such as IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Elsevier. Search terms included "IoT anomaly detection," "ML-based security monitoring," "Edge computing security," "Explainable AI in anomaly detection," and "cloud security in IoT." The focus was on peer-reviewed journal articles, conference papers, and white papers published from 2018 to 2023.

The study's inclusion criteria centered on research addressing (i) ML-based anomaly detection techniques in IoT settings; (ii) security

Table 1
Types of anomalies in IoT-Edge-Cloud systems.

Anomaly type	Examples	Potential consequences
Security-Based	Unauthorized access, DDoS attacks, Data exfiltration	Data breaches, system hijacking, privacy violations
Performance-Based	Excessive latency, packet loss, network congestion	Delayed response times, reduced reliability
Sensor-Based	Faulty sensor readings, missing data, calibration errors	Incorrect ML-based predictions, unreliable decisions

vulnerabilities across IoT, Edge, and cloud computing; (iii) challenges in real-time data processing within distributed IoT architectures; and (iv) the incorporation of XAI for enhanced security in anomaly detection. To maintain a targeted and relevant discussion, studies not explicitly addressing security, anomaly detection, or ML-based methods were excluded. The literature review also provides a comparative analysis of both conventional and modern approaches to anomaly detection, highlighting the shortcomings of current methods and emphasizing the necessity for the proposed multilayered security framework.

2.2. Integration of Internet of Things and cloud computing in data management

The convergence of IoT, CC, and EC has revolutionized data management practices through technological progress. IoT facilitates the interconnection of devices, enabling them to produce, gather, and send vast amounts of data across diverse sectors such as smart cities, healthcare, and industrial automation (Singh, 2019). Nevertheless, efficient data handling necessitates robust storage, processing, and security systems, which traditional standalone IoT setups have found challenging to deliver. The merger of IoT with CC tackles these issues by providing scalable storage, substantial computational power, and centralized data administration (Fortino et al., 2021).

CC delivers Internet-based services, including servers, storage, databases, and analytics, addressing numerous IoT-related challenges. In addition to benefits such as increased geographic reach, reduced infrastructure costs, and widespread accessibility (Kumar et al., 2018), CC can address concerns about IoT energy consumption and data management. However, it is crucial to note that issues related to data transfer, protection, privacy, availability, and security remain (Potluri et al., 2020).

Each IoT device is assigned a unique identifier, facilitating data transmission, collection, and differentiation through the Internet (Sawalha & Al-Naymat, 2021). Moreover, these smart devices can gather, examine and store various types of information (Cheng et al., 2021). However, extensive data generation through network technologies such as 3/4/5G, ZigBee, Wi-Fi, Bluetooth, and wired communication presents significant challenges to Database Management Systems (DBMS), particularly in storing and manipulating heterogeneous data fields (Eyada et al., 2020).

The interaction between IoT, EC, and CC, illustrated in Fig. 1, helps overcome hardware constraints by reducing power consumption and managing big data (Kumar et al., 2018). However, they also face

security vulnerabilities. The authors (Desarda et al., 2020) highlighted that numerous major companies and individuals depend on CC platforms for convenient and maintenance-free storage and data access. However, this symbiotic relationship also has several limitations. Security vulnerabilities persist, and efficient big data management remains a significant challenge. Ensuring data exchange between devices while maintaining confidentiality and security is problematic (Puliafito et al., 2015). Furthermore, an increase in generated data can lead to redundancy and duplicated records, resulting in larger cloud storage requirements (Elbanoby et al., 2019).

Several researchers have proposed solutions to address these issues. A framework that filters and combines data, thus reducing the amount of IoT records stored in a database, was presented by Elbanoby et al. (2019). This system incorporates a Feedback Control Mechanism (FCM) to conserve power and decrease network traffic. In contrast, Nasser et al. (2021) proposed a data scheduling approach that uses a multilayer, priority-based, dynamic, and time-sensitive data processing mechanism (MPDTS), which showed improved performance in terms of latency and data processing speed.

Although IoT provides substantial benefits for managing IoT data, concerns regarding security, latency, and bandwidth constraints continue to pose significant challenges. These issues have prompted the development of alternative architectures like Edge and Fog Computing, which aim to improve real-time data processing and enhance security in IoT environments.

2.3. Edge/Fog computing: an alternative to cloud computing

CC has experienced a surge in popularity due to its ability to facilitate data management on cloud servers. This has met the growing demand for digital services and communications. However, promising advancements in 5G and IoT (IoT) technologies pose challenges to traditional CC. Bandwidth limitations, high latency, and resource allocation challenges make traditional CC suboptimal (Parikh et al., 2019).

In response to these challenges, EC has emerged as a promising alternative. Fig. 2 shows that EC allows data processing using local storage, resulting in real-time computation with significantly reduced latency (Sharma & Sharma, 2019). Moreover, EC is advantageous for IoT devices because it allows data generation and storage in local storage without the need for data transfer to the cloud (Husain & Askar, 2021). Despite these advantages, EC face their own obstacles, such as insecure communication, recovery and backup issues, and limited network visibility (Husain & Askar, 2021).

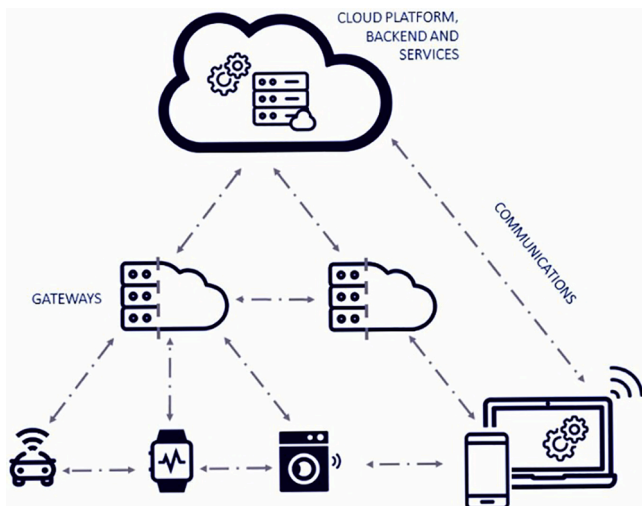


Fig. 1. The interaction between IoT and CC.

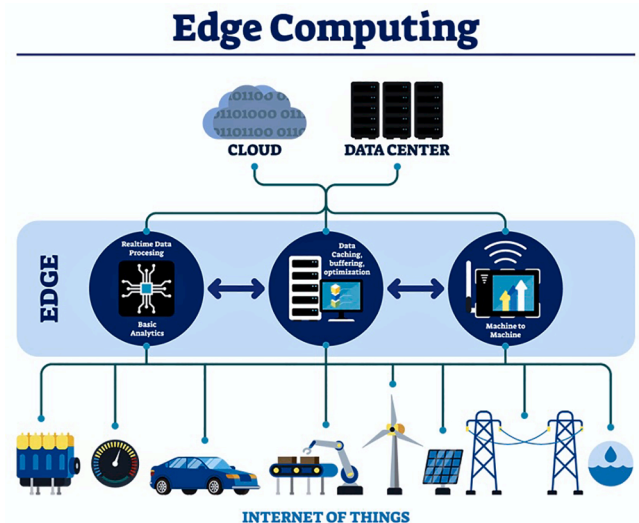


Fig. 2. The general architecture of IoT and EC.

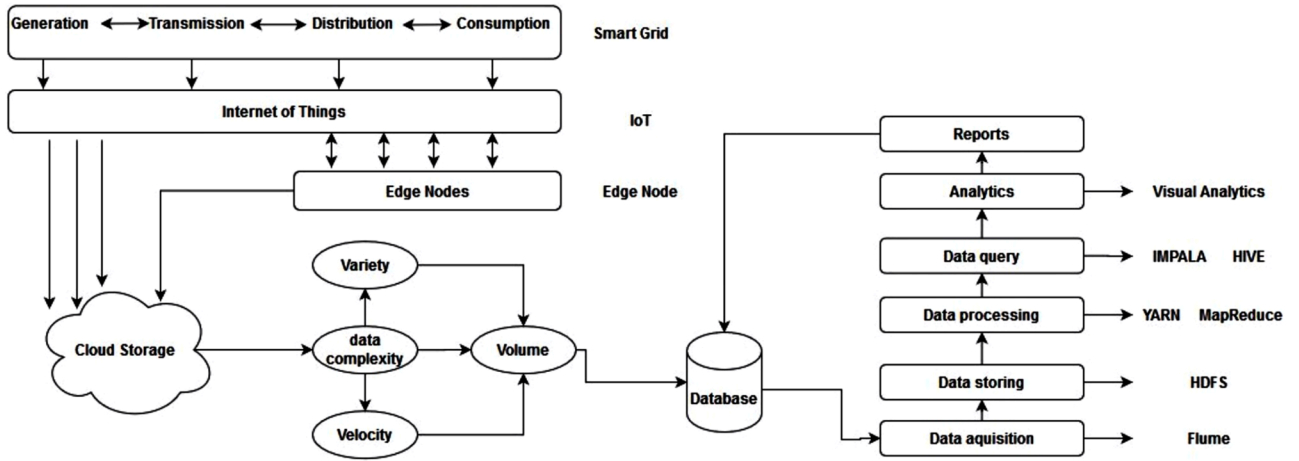


Fig. 3. The use of smart grid information in IoT and EC.

EC is particularly prevalent in the IoT, automotive and manufacturing industries of healthcare (Husain & Askar, 2021). It has applications in data analytics, real-time monitoring, and network management (Parikh et al., 2019). The proposed solution, presented in Fig. 3, utilizes Smart Grid Information (SGI) in IoT and EC systems. This suggests that EC is an efficient method for collecting large-scale data (Ullah et al., 2021). The authors asserted that the generation of heterogeneous devices results in the accumulation of structured, semi-structured, and unstructured data in four distinct stages: Generation, transmission, distribution, and consumption of SGI.

Lopez-Viana et al. (2022) further investigated the interaction between EC and IoT, proposing the utilization of Continuous Integration (CI) and Continuous Deployment (CD) employing GitOps. The authors asserted that GitOps is predominantly used in Kubernetes, but existing methodologies are suitable for IoT EC, as shown in Fig. 4. They discovered that while EC offers apparent advantages, characteristics such as network limitations and device heterogeneity pose challenges during testing and deployment. Furthermore, Parikh et al. (2019) highlighted security and data management vulnerabilities that could potentially compromise systems against cyberattacks. Most EC studies have neglected data management and security considerations. Consequently,

the EC needs to identify the appropriate database type and methods to address the aforementioned challenges.

FC introduces a middle layer between IoT data sources and the cloud, functioning as a technology for data management, storage, and computation, while EC operates on device-level gateways. FC comprises high-performance physical machines with distributed fog nodes creating a decentralized infrastructure (Bouachir et al., 2020).

The author provides a thorough examination of Fog Architecture and its implementation in the Industrial Internet of Things (IIoT) within smart industries. In Bouachir et al. (2020), the author describes the three layers of Fog Architecture highlighting the benefits of incorporating Fog/EC into IIoT systems. These advantages include substantial computing power, improved latency efficiency, improved system capability, greater network flexibility and availability, and optimized data traffic and overhead management.

Furthermore, Bouachir et al. (2020) points out the challenges to consider when designing an edge/fog-based IIoT system, such as the diversity of fog nodes used. The author stresses the importance of implementing an efficient and secure fog infrastructure to ensure smooth communication between fog nodes and IoT devices. The significance of addressing authentication, data integrity, and protection against

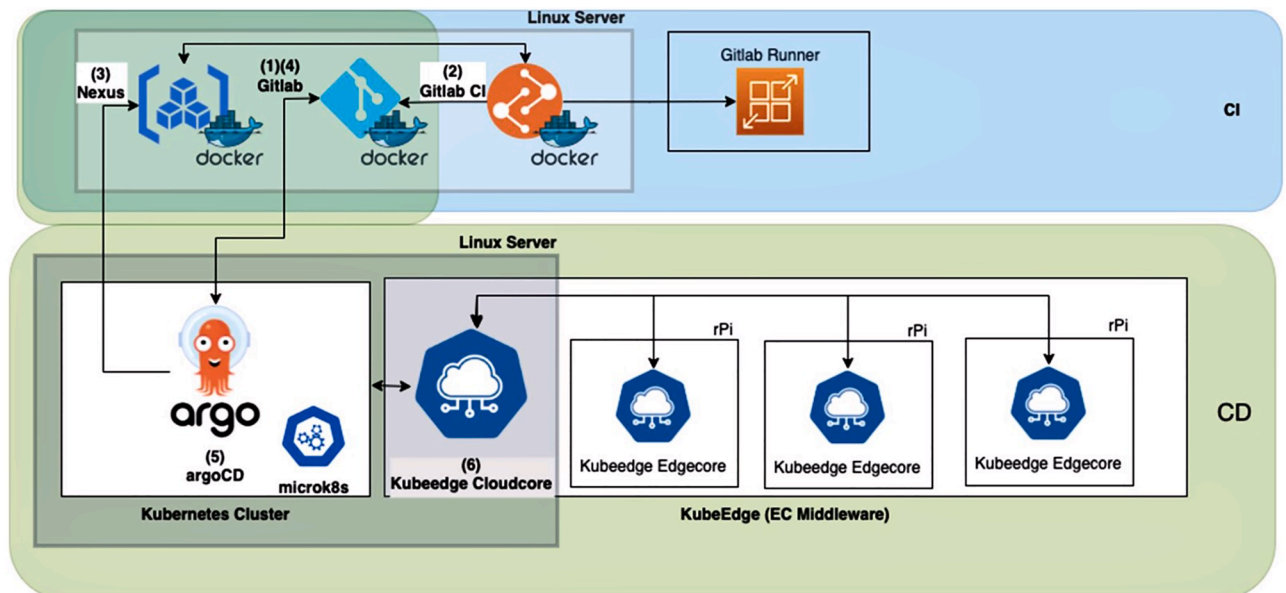


Fig. 4. Continuous integration and continuous development in IoT and EC.

malicious attacks (e.g., denial of service and man-in-the-middle attacks) on fog systems is also emphasized. FC provides numerous benefits, including lower latency, mobility support, and real-time interaction (Alam, 2019; Parikh et al., 2019). However, challenges remain, particularly with regard to security vulnerabilities that can expose systems to interception, hijacking, and other cyberattacks (Parikh et al., 2019).

In conclusion, while CC, EC, and FC offer viable alternatives to conventional cloud-based IoT architectures for data management and IoT integration, they also present unique challenges that need to be addressed. These frameworks require strong security mechanisms and efficient data management techniques to be fully effective. This has led to the integration of ML models for anomaly detection in IoT systems.

2.4. Integration of machine learning, IoT, and cloud computing for anomaly detection

In this subsection, we discuss machine learning techniques for anomaly detection and XAI methods for IoT anomaly detection.

2.4.1. Machine learning techniques for anomaly detection

In the last ten years, ML approaches to detect anomalies in IoT environments have advanced considerably, driven by the increasing complexity and quantity of data produced by IoT devices (Ismaeel & Elmedany, 2022). Various ML techniques, including clustering, classification, and neural networks, have been utilized to identify anomalies with varying levels of effectiveness.

Studies have explored both unsupervised and supervised learning algorithms for real-time anomaly detection (Fortino et al., 2021). Unsupervised learning methods, such as clustering and density-based algorithms, are frequently used in IoT anomaly detection due to their ability to function without labeled training data (Sawalha & Al-Naymat, 2021). On the other hand, supervised learning algorithms depend on labeled data and are mainly used for binary classification tasks (Ismaeel & Elmedany, 2022).

Semi-supervised learning algorithms merge unsupervised and supervised techniques, which is advantageous when labeled data are scarce (Li et al., 2020). Furthermore, Mithran and Gopi (2022) implemented a combination of clustering and classification algorithms to detect anomalies in the IoT sensor data. In Haji and Ameen (2021), a thorough review of appropriate ML algorithms was performed for anomaly detection in IoT networks.

In recent advancements, Li et al. (2021) presented a novel approach called similarity measured isolation forest (SM-IF) for anomaly detection in machine-monitoring data. Although it demonstrates enhanced detection accuracy and computational efficiency, it may face challenges with highly irregularly distributed data sets. In contrast, Koren et al. (2023) proposed a versatile and customizable procedure for anomaly detection, which might require substantial resources for system-specific implementation, making it unsuitable for highly complex or dynamic data systems. Dang et al. (2021) introduced OpenK, a scalable and elastic data cleansing system that employs a cluster-based anomaly detection approach. However, its setup and maintenance may be resource intensive and ill-suited for real-time anomaly detection applications. The study conducted by Kansal et al. (2024) proposed a framework that combines edge computing and federated learning, emphasizing both the privacy-preserving and latency-reducing benefits of offloading computational tasks to local edge devices. The effectiveness of the proposed approach is heavily based on the capabilities of the edge devices. If edge devices lack sufficient computational power or resources, the performance of anomaly detection algorithms may be compromised. The authors Xiang et al. (2024) explained a three-stage framework using federated learning with isolation forest algorithm to achieve superior detection accuracy and robustness, but also enhance data privacy and reduce computational burdens, making it a promising solution for anomaly detection in the continuum of the edge of IoT.

2.4.2. Explainable AI for IoT anomaly detection

Although ML models achieve high accuracy, their black-box nature poses challenges in IoT security applications. To address the comprehensibility of such models, XAI has emerged as a promising research domain that focuses on improving the transparency, interpretability, and reliability of AI systems (Mahajan et al., 2021). Although simpler models may lack the effectiveness of complex models (Dwivedi et al., 2023), XAI techniques, such as Local Interpretable Model-Agnostic Explanations (LIME) and SHAP, have been introduced to improve model interpretability without compromising performance (Gramegna & Giudici, 2021).

- SHAP: Assigns feature importance values, making it useful for understanding which sensor readings contribute most to anomalies.
- LIME: Generates local approximations of model decisions, allowing users to validate ML outputs in real-time (Gramegna & Giudici, 2021).

A key finding is the limited incorporation of XAI in the detection of IoT anomalies, with most research focusing on single ML algorithms to detect anomalies, neglecting the potential advantages of combining multiple data sources or using CC. The synergy between IoT and CC can greatly improve ML-based anomaly detection, facilitating real-time identification of unusual behavior and enhancing system security. This combination also addresses challenges like the lack of labeled data and the requirement for real-time processing. However, integrating IoT and CC requires careful consideration of security, privacy, network stability, and complexity issues. For example, vulnerable IoT devices may lead to security breaches, while unreliable networks can cause data loss or system failures. Furthermore, the integration process can be complex and resource-demanding, necessitating specialized expertise for effective implementation and maintenance.

Although extensive research has been conducted on ML-based anomaly detection and the integration of IoT and CC in data management, there is a significant gap in the literature on the synthesis of these fields. Few studies have explored the potential of ML-based anomaly detection in an integrated IoT cloud environment, and even fewer have provided comprehensive empirical evidence of such systems' performance. Considering the rapid increase in IoT devices and the growing dependence on CC for data storage and processing, this gap represents a missed opportunity. To address this literature gap, this research aims to investigate the unexplored area of integrating ML-based anomaly detection with IoT and CC technologies in data management systems. Combining these domains has the potential to create a highly efficient, scalable, and accurate anomaly detection system, offering significant benefits to data security. This study seeks to make substantial contributions to the applications of both academia and industry through empirical research and a thorough analysis of the results.

3. Research methodology

In this section, we first introduce an overview of our proposed architecture. We then detail the design and implementation of the architecture. Additionally, we describe how anomaly detection and interpretation are performed using XAI within our framework. We also discuss the security measures and compliance considerations, and briefly explain how a voice assistant is integrated into the SSPA.

3.1. Overview of the proposed architecture

This section describes the design and implementation of a ML-based anomaly detection system that incorporates IoT, Edge, and IoT technologies. The suggested method employs a multilayered strategy to identify anomalies in real time, ensure data security, and incorporate XAI. The study adopts an experimental approach, developing a layered ML-based anomaly detection system comprising:

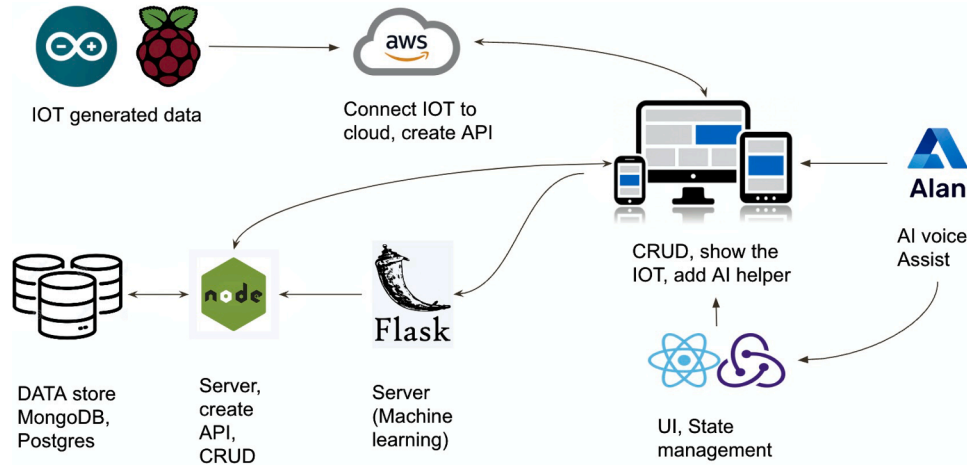


Fig. 5. The proposed ML-based IoT and CC architecture for anomaly detection.

1. IoT Layer: Data is collected from IoT devices and transmitted via MQTT protocol.
2. Edge Layer: Performs data pre-processing, feature extraction, and real-time anomaly detection to reduce cloud dependency.
3. Cloud Layer: Runs ML models (Isolation Forest, Local Outlier Factor) for high-precision anomaly classification.
4. SSPA: Provides real-time visualization, anomaly alerts, and secure data monitoring within an IoT-Edge-Cloud ecosystem. Enhancing transparency in anomaly detection decisions.

Fig. 5 showcases the overall architecture of the proposed system, which leverages various technologies. The process begins with a comprehensive collection of real-time data from IoT devices. These data undergo processing and storage in the cloud, followed by the application of an ML algorithm for anomaly detection. Thorough performance evaluations of the integrated system were conducted, leading to iterative model refinement. The fusion of IoT and CC technologies is crucial to the design's success, as it combines the real-time data collection capabilities of IoT sensors with scalable storage, processing, and analytical features in data management, thus enhancing security and anomaly detection. This integration offers benefits such as real-time monitoring and remote access. In addition, EC minimizes latency and ensures swift, accurate data processing. Moreover, the combination of IoT and CC technologies is particularly advantageous in industrial systems, where relying solely on cloud-based computing may prove inefficient. The subsequent sections provide a more detailed explanation of each subcomponent.

3.2. Design and implementation of our proposed architecture

In this subsection, we present each layer, starting with the IoT Layer (which includes IoT microcontrollers and sensors), the Edge Layer (which focuses on data storage and processing in edge computing), the Cloud Layer (which integrates machine learning algorithms for anomaly detection), and the SPA Architecture (which encompasses design and development). Additionally, we cover data visualization and user interaction, as well as cloud connectivity and secure data transmission.

3.2.1. IoT Layer: IoT microcontrollers and sensors

The development of the proposed system involved a meticulous process to select the appropriate IoT microcontrollers and sensors capable of precise real-time data capture and transmission. While various microcontroller options were evaluated, including popular platforms such as Arduino and Raspberry Pi, the ESP32 was ultimately chosen due to its superior features. These include integrated Wi-Fi capabilities, cost effectiveness, power efficiency, and flexibility for IoT applications, making it ideal for real-time data collection in environments with limited resources.

The programming of the ESP32 microcontroller was performed using the Arduino Integrated Development Environment (IDE), selected for its user-friendly interface and robust community support. To ensure accurate sensor readings and efficient data transmission, specific libraries were employed to facilitate data encoding in JavaScript Object Notation (JSON) format. JSON was preferred due to its compact structure and compatibility with cloud-based applications, which allows smooth data exchange between IoT devices, EC nodes, and cloud storage systems.

A crucial aspect of this research is the combination of IoT and CC technologies. IoT-enabled data acquisition encompasses a wide range of environmental sensors, smart devices, and wearable technologies, ensuring comprehensive and multifaceted data collection. The dataset comprises various parameters, including humidity levels, light intensity, smoke detection, LED display outputs, and motor sensor readings, providing a realistic representation of operational scenarios. Although diverse heterogeneous sensors can be utilized, this study specifically employs low-power, high-accuracy sensors to enhance the reliability of data-driven anomaly detection.

To facilitate scalability and real-time processing, all data collected are securely stored in a cloud-based infrastructure, allowing efficient data retrieval, analysis, and ML anomaly detection. This cloud-centric approach ensures that IoT-generated data are processed, analyzed, and visualized in a secure and scalable environment, thereby enabling effective anomaly detection and predictive analytics in IoT-edge-cloud ecosystems.

3.2.2. Edge layer: data storage & processing in edge computing

This research employs a IoT Infrastructure as a Service (IaaS) model, providing greater flexibility and resource management. Amazon Web Services (AWS) is utilized to enable scalable storage, instantaneous data processing, and comprehensive analysis. To protect data during transfer and storage, we implemented advanced encryption methods and secure transmission protocols.

The AWS IoT Core service is used for cost-effective and effective data handling, enhancing system scalability by managing and processing large datasets. Additionally, Amazon Cloud for IoT is used to securely store and administer high-volume data generated by IoT devices. A cloud-based application is developed to ensure smooth data processing, efficient storage management, and improved security for all IoT-generated information.

Initial data processing and filtering take place at the Edge layer before cloud transmission, enhancing efficiency and reducing unnecessary data transfer. A crucial aspect of this process involves feature extraction, which removes superfluous or irrelevant information, such as eliminating outlier peaks from noisy sensor data. In addition, a threshold

anomaly detection system based on statistical analysis is used, allowing immediate identification of irregularities. For example, when a temperature reading exceeds 100°C, an alert is triggered. To further enhance system reliability, the Edge layer incorporates local storage capabilities, enabling temporary data retention during network outages. This ensures that once connectivity is reestablished, the stored information can be transmitted without loss.

Each IoT device in the AWS IoT console receives a unique identifier and crucial security credentials, such as digital certificates and cryptographic keys. This allocation is essential for establishing secure, authenticated communication between IoT devices and the AWS IoT platform. The MQTT protocol is designed to enable secure and efficient communication between AWS IoT services and IoT-based microcontrollers. This configuration allows for low-latency data transmission, ensuring reliable message exchange between edge devices and cloud infrastructure. Through this integrated cloud-edge computing approach, our methodology improves data security, processing efficiency, and real-time anomaly detection in IoT-driven environments.

3.2.3. Cloud layer: machine learning algorithms for anomaly detection

The SSPA incorporates an anomaly detection system that utilizes both supervised and unsupervised ML methods. For the supervised component, the system employs threshold-based detection, while the unsupervised aspect utilizes an Isolation Forest algorithm. These methodologies were chosen because of their proven efficacy in prior research and their capacity to handle large-scale, high-dimensional data sets typical of IoT environments.

Local outlier factor algorithm (threshold-based)

Statistical analysis employing threshold-based anomaly detection is a common method for identifying data points that significantly deviate from the central tendency of a data set. This technique calculates thresholds using the mean and standard deviation of the data.

The proposed system incorporates a React-based SSPA with a dynamic threshold-based anomaly detection mechanism that adapts to real-time data streams. The process involves:

1. Data Collection: A function receives and collects the incoming data.
2. Real-time Analysis: Custom hooks within React's state management system perform immediate calculations.
3. Threshold Calculation: The computed values are saved in the Redux Store for future analysis.

React's custom hooks are essential for computing statistical measures like minimum, maximum, mean, standard deviation, and threshold values from incoming IoT sensor data. These calculations are integrated into the functions *useState* and *useEffect* to ensure efficient data management and real-time responsiveness.

Accurate threshold estimation is crucial for anomaly detection. The process involves defining a parameter k , which is multiplied by the standard deviation and added to the mean of the data set. This can be expressed as $T = \mu + k\sigma$, where T represents the threshold, μ is the mean, k is a constant, and σ denotes the standard deviation.

The value of k greatly affects the sensitivity of anomaly detection. Higher k values raise the threshold, reducing false positives but potentially overlooking true anomalies. In contrast, lower k values decrease the threshold, increasing sensitivity but risking more false positives.

The *useThreshold* custom hook in React efficiently implements this threshold calculation. It monitors changes in the input data array and recalculates the mean and standard deviation when new data arrive. The computed threshold dynamically updates the state, ensuring that the anomaly detection system adapts in real-time.

Additional custom hooks, such as *useFilteredIoTsAverageValue* and *useFilteredIoTsStdValue*, enable specialized statistical operations, including data filtering prior to applying threshold-based anomaly detection. This modular design improves the scalability and reusability of IoT applications.

Isolation Forest algorithm

The Isolation Forest (iForest) algorithm, developed by Liu, Ting, and Zhou (2008), is a popular unsupervised method to detect anomalies. This approach is based on the premise that outliers are scarce and distinct, exhibit unique feature distributions, and occur infrequently within datasets.

iForest functions by repeatedly dividing the data space using randomly chosen features and split values between the minimum and maximum feature values. This process creates a binary tree structure known as an isolation tree (iTree). Isolation occurs when partitioning results in a single instance at a node. The key advantage of this technique lies in the length of the path from the root to the terminating node. The average path length across a collection of iTrees serves as a measure of normality and is used to calculate an anomaly score. Anomalies are characterized by shorter path lengths as they are easier to isolate.

Mathematically, the anomaly score $s(x, n)$ of instance x in an iTree of n instances is defined as

$$s(x, n) = 2^{(-E(n)/c(n))} \quad (1)$$

where $E(n)$ is the average path length of instance x in iTrees. $c(n)$ is the average path length of an unsuccessful search in a Binary Search Tree (BST) given n external nodes, and can be approximated as

$$c(n) = 2H(n-1) - (2(n-1)/n) \quad (2)$$

where $H(i)$ denotes the harmonic number. This can be estimated by $\ln(i) + 0.5772156649$ (Euler's constant).

If s is close to 1, then the instance is an anomaly; if s is smaller than 0.5, then the instance is considered a normal instance; if all instances have s approximately equal to 0.5, then the entire sample does not have any distinct anomaly.

The iForest method is particularly well suited for high-dimensional IoT datasets due to its low computational requirements, scalability, and ability to operate without labeled training data. These attributes make it an excellent choice for real-time anomaly detection in distributed IoT environments.

3.2.4. SSPA architecture: design and development

The SSPA architecture serves as the interactive and computational interface that facilitates real-time data processing, anomaly alerts, and visualization within the proposed IoT-Edge-Cloud architecture. The architecture is designed to seamlessly connect IoT microcontrollers, EC devices, and cloud-based ML algorithms, offering a secure and efficient method for gathering, storing, and examining data. Using cutting-edge web technologies such as React.js and Redux, the SSPA architecture delivers a user experience that is scalable, state managed, and highly interactive, as shown in Fig. 6. Furthermore, it employs AWS, WebSockets, and MQTT protocols to ensure secure data transmission with minimal latency across the IoT, edge, and cloud layers. This portion of the text provides an in-depth exploration of the SSPA's design, structure, elements, cloud connectivity, and its integration with ML models for identifying anomalies.

Architectural Design and Core Technologies The SSPA employs a modular and component-based design to boost scalability, ease of maintenance, and efficiency. React.js is selected as the framework due to its proficiency in handling dynamic user interfaces, utilizing its Virtual DOM to enhance rendering speed by updating only necessary components instead of reloading the entire page. This approach improves responsiveness, particularly for real-time IoT data visualization. Furthermore, Redux is utilized as a state management tool to ensure centralized and synchronized data flow among various application components. Redux's capability to efficiently process real-time sensor data updates makes it well-suited for IoT-driven applications that require consistent handling of data streams from multiple sources.

The application's core structure is built upon two essential files: *index.js*, which initializes the application, and *App.js*, which establishes the main layout and framework of the SSPA. The interface comprises

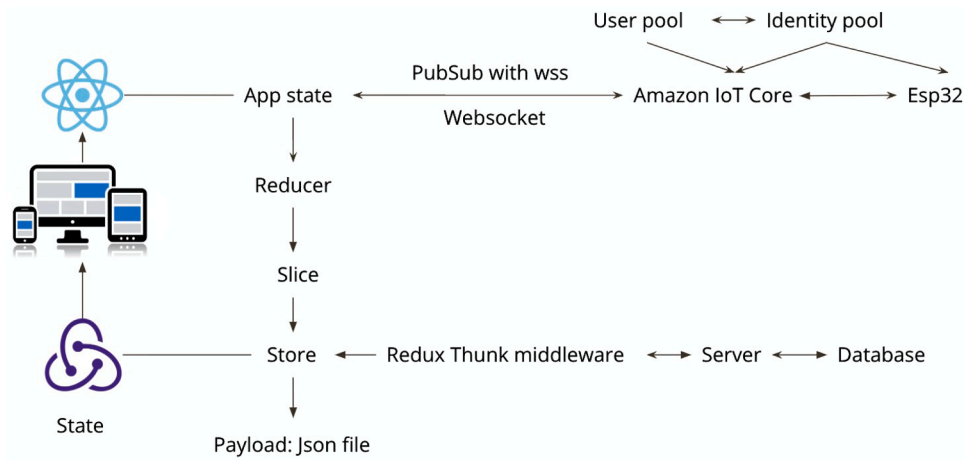


Fig. 6. SSPA: AWS cloud and server communication.

six independent functional components, each dedicated to specific tasks such as data entry, removal, retrieval, and real-time anomaly identification. This modular structure improves code reusability, simplifies debugging, and facilitates future system expansion.

3.2.5. Data visualization and user interaction

The SSPA's primary aim is to provide user-friendly and dynamic data visualization, enhancing IoT data management and anomaly detection. Developed with React-Bootstrap, the Table Page Component acts as the central interface for showcasing, sifting through, and examining IoT-generated information. This element ensures adaptable UI design, enabling access across various devices, from computers to mobile phones.

The Table Page Component's core features include instantaneous data filtering, page navigation, and arrangement, allowing users to efficiently explore and evaluate extensive IoT datasets. The component incorporates WebSocket technology for real-time data updates, ensuring the displayed information remains up-to-date as new sensor readings are received. In addition, search and filtering tools enable users to locate specific records based on predetermined criteria, improving data analysis efficiency. The integration of a color-coded alert mechanism instantly highlights anomalies detected in the dataset, improving user awareness of potential issues in IoT sensor readings.

3.2.6. Cloud connectivity and secure data transmission

The SSPA facilitates a secure and streamlined connection between IoT devices, EC components, and cloud infrastructure through AWS.

The system incorporates AWS Cognito authentication to regulate user verification and access control, ensuring that system interactions are restricted to authorized personnel. Cognito delivers two primary authentication functionalities: User Pools for authenticating users and managing directory services, and Identity Pools for creating temporary credentials to access AWS resources. This approach improves system security by reducing the exposure to unauthorized access and data breaches.

To enable real-time, low-latency data exchange, the SSPA leverages WebSocket communication and integrates the MQTT protocol. WebSockets establish an ongoing, bidirectional communication channel between the SSPA and cloud servers, facilitating immediate data updates. Simultaneously, the MQTT protocol is utilized for efficient, lightweight communication between IoT microcontrollers and AWS IoT Core, minimizing bandwidth requirements. The fusion of these technologies yields a highly responsive and secure data transmission framework, optimizing the processing of real-time IoT data.

Local server architecture and database integration

A Node.js-based local server serves as an intermediary layer between the SSPA and cloud infrastructure to efficiently handle IoT data. This server performs data preprocessing, storage, and anomaly detection before sending relevant information to the cloud. Node.js was selected as the development framework due to its event-driven design and nonblocking I/O operations, which improve the system's overall responsiveness. The server architecture, as illustrated in Fig. 7, follows a

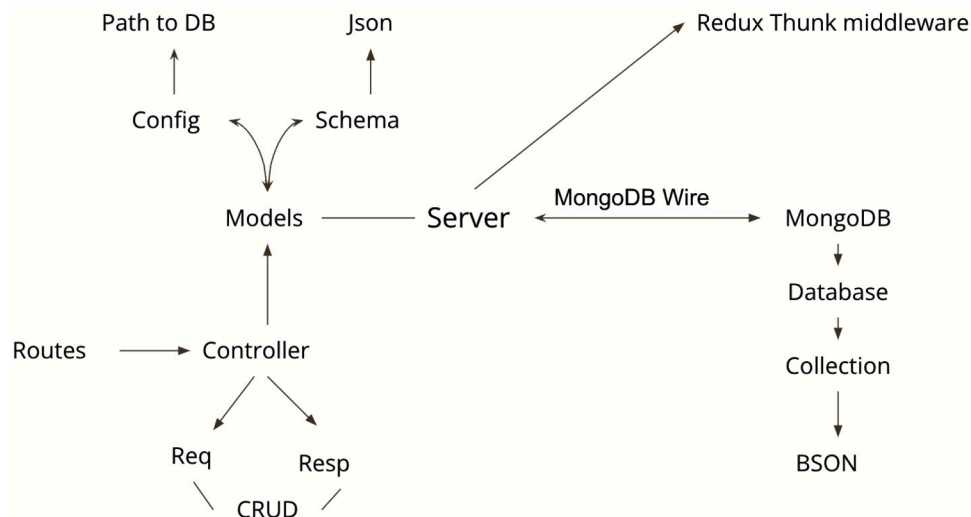


Fig. 7. Node.js local server architecture.

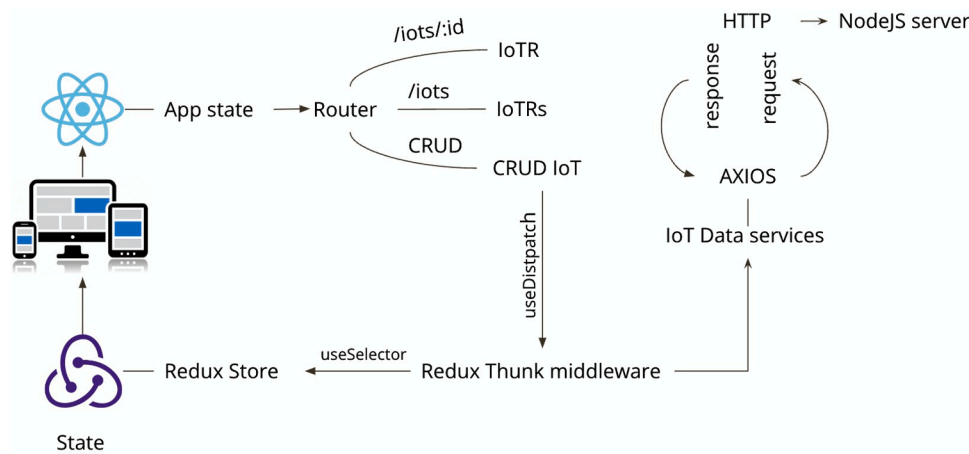


Fig. 8. Integration of SSPA into local server.

modular design, with `server.js` functioning as the main request handler and a structured directory system that organizes configurations, controllers, models, and API routes.

The system incorporates a MongoDB database as its primary data storage solution, selected for its ability to scale, adapt, and manage high-volume IoT data streams (Eyada et al., 2020). To ensure efficient and secure data transfer between the SSPA and the database, the MongoDB Wire Protocol is utilized. This database configuration enables real-time querying, data indexing, and smooth retrieval, facilitating effective analysis of IoT data.

Integration of redux toolkit for state management

The Redux Toolkit serves as the main state management system to improve the synchronization of real-time data between SSPA, local server, and cloud infrastructure, as shown in Fig. 8. This implementation comprises three essential elements: a Service Module that manages CRUD operations using the Axios library; a Redux Slice for handling asynchronous API requests and state modifications; and a Redux Store Configuration that consolidates all state management functions. This well-organized approach facilitates smooth real-time data updates, event-driven interactions, and improved application efficiency.

3.3. Anomaly detection and interpretation using explainable AI

The SSPA connects to cloud-based ML models on a Flask server to perform real-time anomaly detection for IoT sensor data, as shown in Fig. 9. The system employs a dual approach for anomaly detection,

combining threshold-based statistical analysis with unsupervised iForest techniques. The threshold method establishes dynamic anomaly limits by calculating the mean and standard deviation of incoming data, while the iForest algorithm identifies anomalies by examining feature deviations within the dataset.

To facilitate seamless communication between the SSPA and the Flask ML server, a RESTful API and WebSocket integration are implemented, enabling two-way data exchange between the front-end interface and ML-based back-end system. This integration adheres to a client-server model where: 1) SSPA requests processed sensor data from the Flask server via an HTTP GET request, 2) the Flask server extracts data from MongoDB, applies ML models, and sends back anomaly predictions as a JSON response, and 3) SSPA exhibits anomalies in real time through dynamic charts, tables, and alert notifications. The Flask API consists of two main routes.

- **Upload:** This route accommodates HTTP POST requests, managing the CSV file upload process. Upon verification of the file format, the CSV file was saved to a designated upload folder, and its data were read using Pandas. The data were then pre-processed, and anomalies were detected and returned as JSON responses.
- **Anomalies:** This route handles HTTP GET requests and returns the detected anomalies. An error message is generated without detected anomalies or if a CSV file has not been uploaded.

Once the data are received by the Flask server, they are processed using an ML-based anomaly detection pipeline. The system selects the

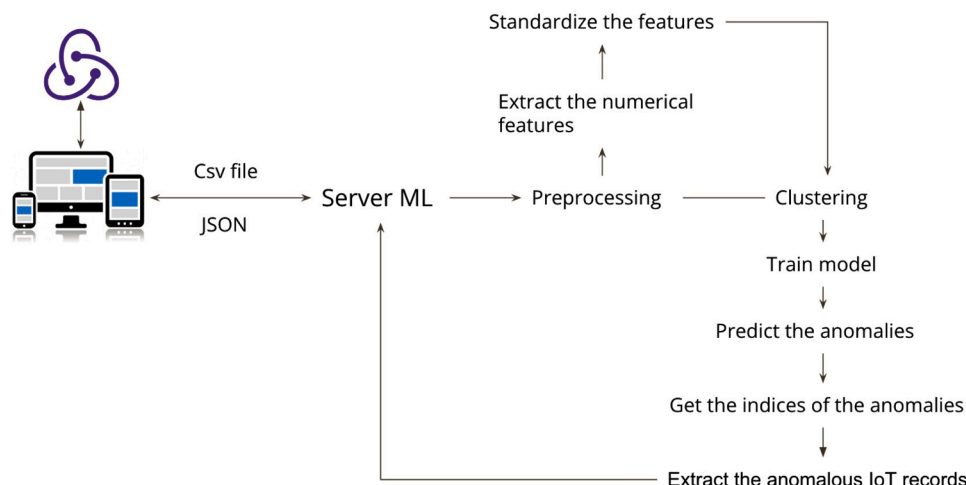


Fig. 9. Integration of SSPA into ML server.

SSPA Tables IoTs IoTs Statistic Add IoTs ML							
Anomalies							
Title	Time	MAC	Humidity	Motor Status	Led Status	Light Intensity	Smoke Detected
ESP32 Sensor Data	2753	F4:89:09:5C:61:6A	98	0	0	5	0
ESP32 Sensor Data	2798	F4:89:09:5C:61:6A	42	0	0	98	0
ESP32 Sensor Data	4894	F4:89:09:5C:61:6A	99	0	0	9	0
ESP32 Sensor Data	4934	F4:89:09:5C:61:6A	42	0	0	99	0
ESP32 Sensor Data	5064	F4:89:09:5C:61:6A	42	0	0	98	0
ESP32 Sensor Data	5559	F4:89:09:5C:61:6A	99	0	0	15	0
ESP32 Sensor Data	147	F4:89:09:5C:61:6A	42	0	0	3	0

Fig. 10. Detected anomalies.

relevant sensor features (e.g., temperature and humidity) for anomaly detection. The features were normalized using Standard Scaler to ensure consistent input ranges. The Flask server applies the iForest and Local Outlier Factor (LOF) algorithms. The anomaly scores were computed for each data point. Threshold-based classification determines whether a sensor reading is normal or abnormal. The anomaly predictions were stored in MongoDB for future analysis. A real-time WebSocket connection sends the detected anomalies directly to the SSPA UI, as shown in Fig. 10.

The SSPA dashboard dynamically displays anomaly predictions received from the Flask server. The interface features a color-coded real-time alert mechanism to highlight irregularities, along with various visualization tools. These include line charts to track sensor trends, scatter plots to identify unusual readings, and a tabular view to list detected anomalies with their corresponding timestamps. To improve understanding, the system employs XAI methods, specifically SHAP. This approach calculates the contribution of each sensor attribute to the anomaly score, offering insights into the factors that influence anomaly classification. The SSPA incorporates SHAP summary plots, enabling users to identify the most influential features in anomaly detection. For example, a classification of anomalies might show that humidity contributed 65 % to the detection decision, allowing users to evaluate the rationale behind the flagged anomalies.

3.4. Security measures and compliance

Ensuring secure data transmission and preventing unauthorized access are crucial aspects of the SSPA-ML integration. The system utilizes Transport Layer Security (TLS/SSL) encryption to protect API communication between the SSPA and the Flask server. Furthermore, AWS Cognito authentication verifies user access for API requests, while Role-Based Access Control (RBAC) limits unauthorized access to anomaly detection endpoints. To preserve data integrity, the system performs checks to ensure that classified anomalies cannot be altered or manipulated post-detection. Table 2 highlights the security mechanisms implemented to mitigate cyber threats.

The proposed SSPA framework successfully combines IoT, Edge, and IoT technologies to create a flexible, protected, and smart system to monitor IoT devices and identify anomalies in real time. Using React, Redux, AWS, WebSockets, and ML techniques, the architecture improves instantaneous performance, safeguards, and data-based decision processes. This system establishes a solid groundwork for immediate and sophisticated IoT applications, making substantial contributions to the study of intelligent environments enabled by IoT.

Table 2

Security measures in across layers.

Layer	Security risk	Countermeasure
IoT Layer	Unauthorized access, malware	TLS encryption, authentication keys
Edge Layer	Data interception, DDoS attacks	Firewall, Intrusion Detection (IDS)
Cloud Layer	API exploitation, data leaks	AES-256 encryption, IAM policies
SSPA	Unauthorized dashboard access	Amazon Cognito authentication

3.5. Integrating a voice assistant into the SSPA

SSPA is an interactive web platform with an adaptive interface. Infusion of a voice assistant into the application, as shown in Fig. 11, promises substantial improvements in user experience by introducing the convenience of voice-driven control over the application. Alan AI, a renowned voice assistant, can seamlessly merge with the SSPA, promoting proficient voice interactions.

To integrate Alan AI into the SSPA React application, it is necessary to embed the Alan AI SDK into the project and establish communication between the voice assistant and the components of the application. Developing an intent script becomes crucial in defining the range of voice commands and actions that the assistant should respond to once the Alan AI is implemented. This script includes codes to identify user intentions, translate speech, and execute associated functions.

Incorporating a voice assistant into an SSPA has numerous benefits. Users can quickly retrieve application-related information via voice commands, enhancing accessibility and simplifying usage. Furthermore, voice-mediated interactions facilitate seamless page routing and navigation, thus improving overall user experience.

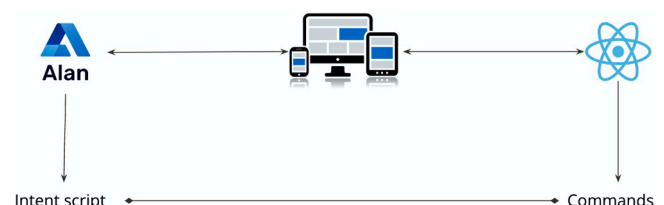


Fig. 11. Integration of voice assistance with SSPA.

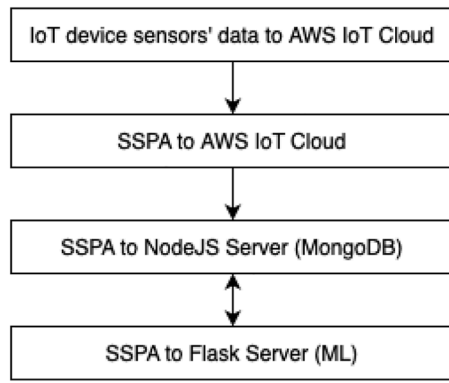


Fig. 12. Components of SSPA in terms of data management.

4. Results and discussion

This study explores the potential of integrating ML, IoT, CC, and EC to improve anomaly detection in data management. The high AUROC scores achieved demonstrate the effectiveness of the proposed ML-based anomaly detection system in accurately identifying anomalies within data streams generated by IoT devices and processed in the cloud. The system's scalability and robustness further highlight its relevance for real-world environments, where efficiently managing large datasets and numerous connected devices is critical for modern digital infrastructure.

4.1. System demonstration and data flow

The integration of the SSPA framework facilitates seamless communication between system components, as shown in Fig. 5. In terms of secure data management, the architecture comprises four primary components, as illustrated in Fig. 12.

The experimental setup involved the deployment of an ESP32 microcontroller selected for its built-in Wi-Fi capability to generate and transmit sensor data to the cloud. The process commenced with programming of the microcontroller to establish Wi-Fi connectivity and initiate data transmission. This interaction was validated using the ARDUINO IDE, in which the cloud interface was used to monitor connection status and visualize generated data. Furthermore, it aids in generating JSON objects consisting of an array and messages, but also includes extrane-

Table 3

Stage: processing time from data generation to detecting anomalies.

Steps	Time
Component-I: IoT-Layer to Edge-Layer	≈ 2 sec
Component-II: SSPA to Edge-Layer	≈ 3 sec
Component-III: SSPA to Node.js Server	Every 10 sec
Component-IV: SSPA to/from Cloud-Layer	≈ 1 sec

ous data. Thus, the JSON objects generated are pivotal for subsequent processes.

IoT devices were configured to collect and transmit data to the cloud, where the initial processing involved the removal of extraneous information to produce a structured data set. The refined dataset, illustrated in Fig. 13, was transmitted in real time to a Node.js server integrated with MongoDB, for efficient data storage and retrieval. This setup enables real-time monitoring within the SSPA framework, facilitating threshold-based anomaly detection, as depicted in Fig. 14. The processed data was then forwarded to the Flask server, where ML algorithms were used for anomaly detection.

In the final phase, the detected anomalies were categorized based on unique message identifiers, with results presented in an intuitive tabular format at the SSPA interface, as shown in Fig. 15. Anomalies were marked as 'true', while normal data points were marked as 'false'. The entire process, from data generation to anomaly detection, demonstrated an average processing time of approximately six seconds, as summarized in Table 3. However, performance may vary due to factors such as network stability, server capacity, and computational efficiency.

4.2. Anomaly detection performance and comparative analysis

The system under consideration utilizes the iForest algorithm, a well-known unsupervised ML method to detect anomalies. To maintain data quality, several preprocessing steps were implemented, including the conversion of categorical variables through label encoding, filling in missing data points with column averages, and adjusting infinite values to normal ranges. These preparatory measures ensure the reliability of the data set and enhance the effectiveness of subsequent analytical procedures.

After preprocessing, the iForest algorithm was applied to detect unusual data points, which were then incorporated into the data set as a new classification column. Data points classified as normal were given a value of (-1), while those identified as anomalous were assigned a

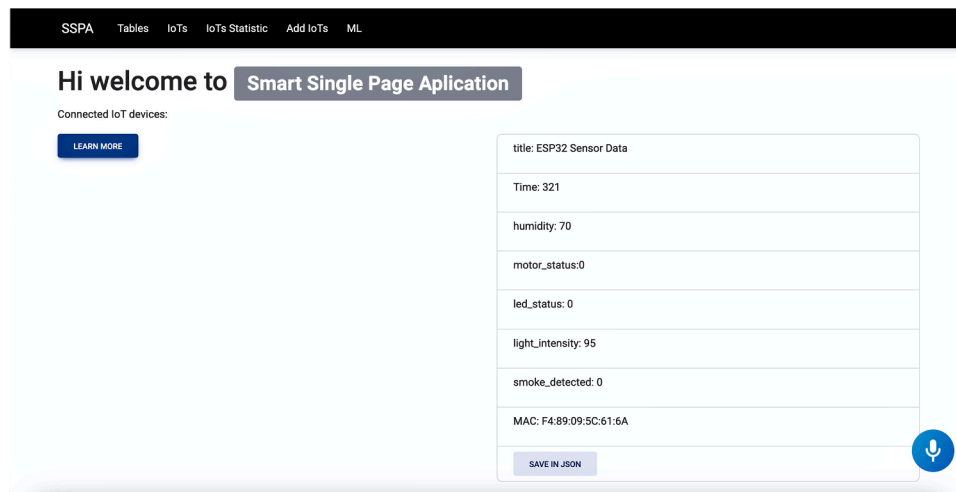


Fig. 13. Main interface of SSPA showing IoT device data, connected with AWS IoT Cloud.

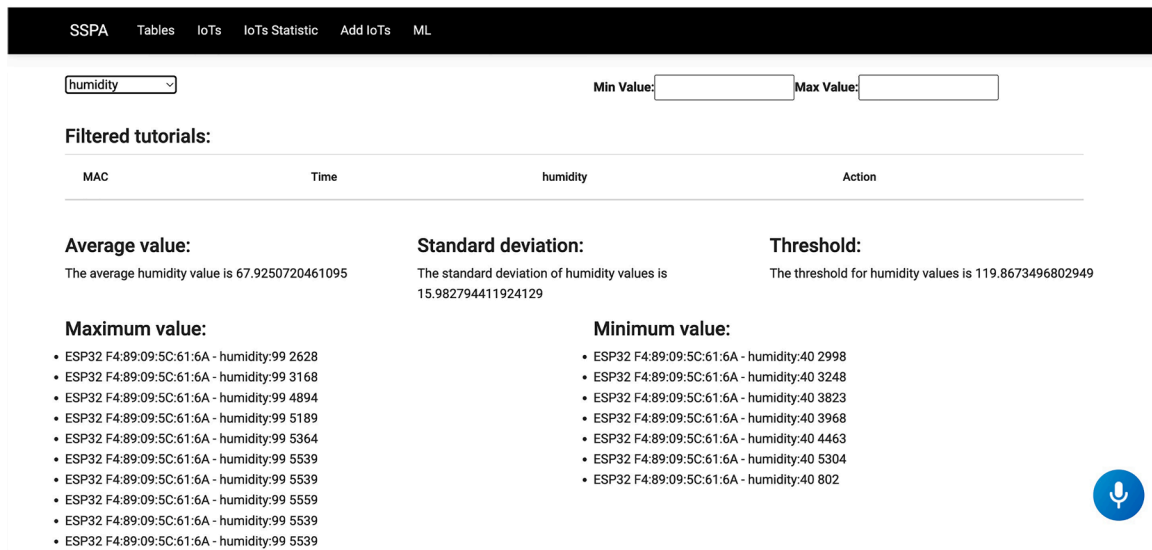


Fig. 14. Data visualization within SSPA.

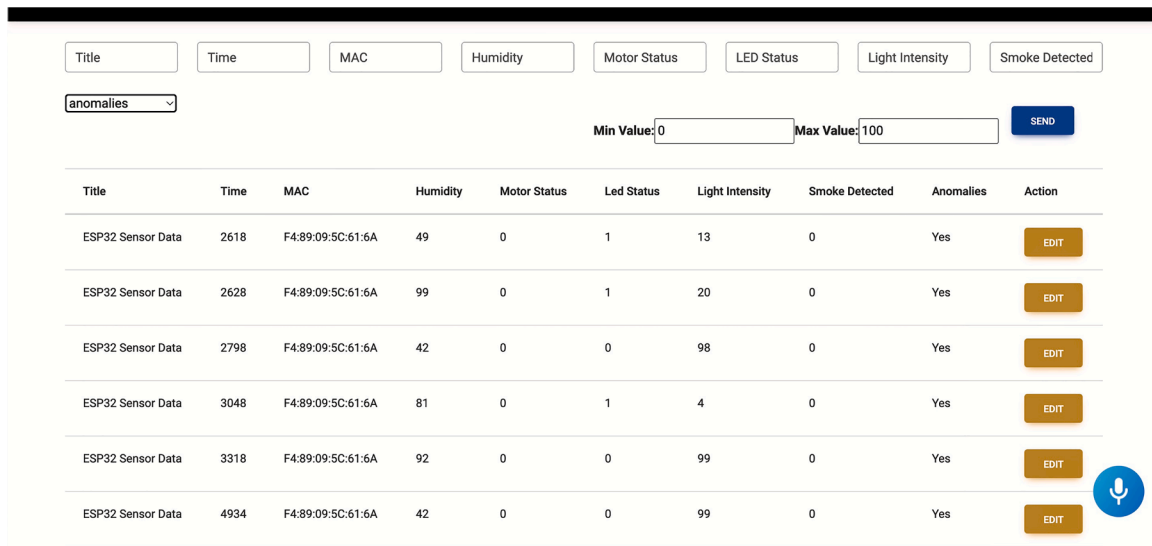


Fig. 15. Detected anomalies using ML algorithms.

Table 4

Benefits and drawbacks of various ML methods for anomaly detection.

Method	Benefits	Drawbacks
Isolation Forest	Efficient for high-dimensional data Robust to noise Handles large dataset	Less effective for irregular distribution Hyper-parameters tuning required
Local Outlier Factor	Detects local anomalies Handles varying densities	Sensitive to hyper-parameters Limited performance in high-dimensions Complexity increases with dataset size
Clustering	Identifies group anomalies Large dataset handling Can be combined with other methods	Assumes anomalies are not in clusters Needs suitable distance metric Sensitive to algorithm choice

value of (1). The resulting data set was then stored for future analysis to improve anomaly detection capabilities.

A comparative study was conducted to evaluate the performance of different anomaly detection algorithms, specifically LOF and iForest, in identifying irregularities within IoT sensor data. The findings presented in Table 4 highlight the relative advantages and drawbacks of each method. The iForest demonstrated high efficiency in processing high-dimensional data and showed resilience to noise. However, its performance can be affected by irregularities in data distribution and requires

fine-tuning of hyperparameters. In contrast, LOF excelled at detecting local anomalies and adapting to varying data densities, but was sensitive to hyperparameter settings and less efficient when dealing with high-dimensional data. Clustering-based approaches have shown promise in identifying group anomalies, although their effectiveness depends on assumptions about the underlying data distribution.

Given the inherent class imbalance in anomaly detection datasets, the AUROC was selected as the primary performance metric. As shown in Table 5, the iForest algorithm achieved an AUROC score of 1.00,

Table 5
Comparison of anomaly detection accuracy.

Method	AUROC Score	Latency	Explainability
Isolation Forest (Proposed)	1.00	28ms	SHAP-based
Local Outlier Factor (LOF)	0.89	35ms	None
Clustering	Not applicable	–	None

outperforming LOF, which achieved a score of 0.89. These results underscore the efficacy of the proposed approach for accurately distinguishing between normal and anomalous data points.

4.3. Explainable AI and interpretability

The proposed framework incorporates XAI methods to improve comprehensibility, allowing users to examine the most unusual characteristics that contribute to identified anomalies. This strategy is consistent with XAI principles and offers insight into the underlying causes of irregular sensor readings. For example, when a sensor shows unexpected changes in humidity, users can explore possible reasons such as sensor defects, environmental factors, or system malfunctions. As shown in Fig. 16, the SSPA interface displays anomaly scores and emphasizes key characteristics that influence anomaly classification.

The anomaly score, which ranges from -1 to 1, measures how much a specific data point deviates from the normal distribution seen in the training dataset. A lower score suggests a greater deviation, indicating a higher probability of an anomaly. The iForest algorithm calculates this score by examining the average length of the path needed to isolate each data point within an ensemble of decision trees. Data points that can be isolated with fewer partitions have shorter path lengths, resulting in higher anomaly scores.

This scoring mechanism enables an effective prioritization of anomalies, allowing users to methodically investigate unusual patterns. In addition, the system offers explanatory tools to help users understand anomaly classifications. These insights facilitate informed decision making and the implementation of preventive measures to avoid recurring anomalies, thus improving the efficiency of IoT systems.

4.4. Security and database management architecture

The proposed architecture implements a security framework for managing databases and facilitating IoT-cloud communication to guarantee safe data transmission and storage. Security measures, including encryption protocols, authentication certificates, and structured access control policies, are described in Tables 6 and 7. For secure IoT-cloud communication, the system employs the MQTT protocol, while the MongoDB Wire Protocol protects database interactions. These security features maintain data integrity and prevent unauthorized access attempts.

A comparative analysis of the security protocols between the proposed SSPA and other IoT-CC architectures, such as Industrial Cyber-Physical Systems (ICPS) with FC, is provided in Table 8. The results indicate that the SSPA architecture offers robust authentication mechanisms, improved encryption standards, and enhanced access-control policies. Furthermore, a comparative evaluation of the data management architectures, as presented in Table 9, highlights the advantages of the SSPA framework over alternative solutions, particularly in terms of cost-effectiveness and scalability.

Based on the comprehensive analysis presented in Table 9, the SGI architecture exhibits significant scalability through the integration of EC and the Hadoop cluster. However, this improved scalability comes with increased costs as a result of the infrastructure requirements of EC and Hadoop. In contrast, SSPA uses cloud services and MongoDB, offering a more cost-effective solution with enhanced scalability. This combination of affordability and adaptability makes SSPA a more attractive option for organizations aiming to manage data within the IoT and CC frameworks. Additionally, SSPA demonstrates superior data analytics capabilities by incorporating a Flask server for ML-based anomaly detection. Although both architectures prioritize data security and maintenance, SSPA provides greater flexibility in implementing robust security measures through its reliance on cloud service providers and server configuration. Taking into account factors such as cost efficiency, scalability, and data analytics capabilities, SSPA emerges as the more optimal solution.

It is important to highlight the significant advantages of using MongoDB as the primary storage solution for effectively managing

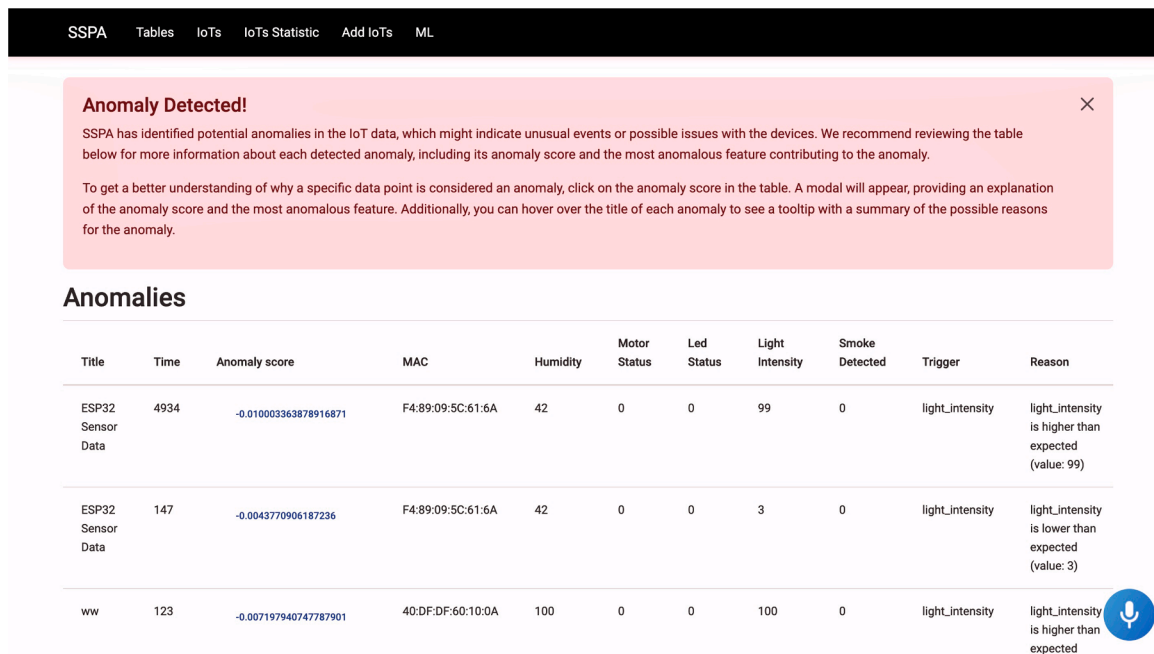


Fig. 16. SSPA anomaly score and most anomalous features.

Table 6
Data management in SSPA.

IoT devices	AWS IoT cloud	Redux	MongoDB	Flask (ML)
Collect data from sensors	Receive sensor data and store in cloud	Fetch from API	–	–
Send data to cloud	Expose API	Update the store	–	–
–	Update data records	Send updates to API	–	–
–	–	Fetch from MongoDB	Store data	–
–	–	Send to Flask	Retrieve/Modify	ML processing
–	–	Store results	–	Send results
–	–	Send results to API	Store results	–

Table 7
Security measures in SSPA and MongoDB.

IoT Devices	AWS IoT Cloud	SSPA	MongoDB
Device Certificate	Policy (rules and permissions)	Identify pool ID (access AWS services)	BSON format, Unique ID
Private key	Rules (process MQTT messages and route to AWS services/devices)	User Pool ID (manage user registration)	Transport Layer Security (TLS/SSL)
Public key	–	MQTT Pool ID (connect to AWS IoT Core)	–
Amazon Root CA1	–	–	–
Amazon Root CA2	–	–	–

Table 8
Security comparison between proposed SSPA and ICPS with Fog architecture.

Security Aspect	SSPA	ICPS and Fog Architecture
Authentication	Identity Pool ID, User Pool ID, MQTT Pool ID	Fog Node Authentication
Data Integrity	TLS/SSL, Amazon Root CA1, Amazon Root CA2	Data integrity in a multilayered processing system
Data Encryption	TLS/SSL for MongoDB	Data encryption at fog nodes and cloud layer
Access Control	AWS policies, rules, and permissions	Access control management at fog nodes and cloud layer
Communication Security	Device Certificate, Private and Public Keys	Secure collaboration and intractability between heterogeneous devices such as Fog-to-Fog, Fog-to-Cloud, and Fog-to-IIoT devices
Malicious Attacks Protection	Close network, a unique ID for each IoT device	Protection against denial of services and man-in-the-middle attacks (do not show)

Table 9
Comparison of SGI and proposed SSPA architecture.

	SGI	Proposed SSPA
Data Flow	IoT devices→edge computing→Hadoop cluster→data analytics tools and visualization	IoT devices→cloud→Node.js server→MongoDB→Flask server→data analytics tools and visualization
Data Storage	Low-cost cloud storage (for data not handled by edge nodes)	MongoDB
Data Processing	Required (inaccurate and incomplete data handled)	Required (inaccurate and incomplete data handled)
Data Acquisition	Flume used for collection, aggregation, and transfer to Hadoop	Amazon IoT core and Node.js server used for acquisition, collection, and transfer to MongoDB and Flask
Data Querying and Analysis	Hive and Impala used	Axios, fetch, and WebSocket were used
Data Analytics	SAMOA used for distributed streaming ML framework	Flask server used for ML anomaly detection
Explainable Intelligence	Not supported	Supported
Data Visualization	Tableau used	React and Bootstrap used
Scalability	Scalable with edge computing and Hadoop cluster	Scalable with cloud and MongoDB
Cost	Higher cost due to edge computing and Hadoop cluster	Lower cost due to cloud and MongoDB
Data Security	Relies on edge computing for security, potential for natural disasters to affect cloud infrastructure	Security measures dependent on cloud provider and server configuration
Maintenance	Requires maintenance and updates for edge computing and Hadoop cluster	Requires maintenance and updates for cloud infrastructure and servers
Support	Only Hadoop services	Web, Cloud, Data, Style services

large-scale datasets. As a NoSQL database, MongoDB offers horizontal scalability, allowing efficient handling of large data volumes and facilitating high read/write rates. Furthermore, MongoDB's document-based storage structure enables flexible management of data from various IoT devices with different data formats. The distribution of responsibilities among distinct architectural components, such as data collection (micro-controller), data storage (MongoDB), and data processing (Flask Server for ML), enhances resource management by allowing independent scaling of each component. This architectural approach ensures a responsive and efficient system even as the volume of data generated by IoT devices increases. To maintain system integrity and security, SSPA implements robust security measures.

Finally, to highlight the improvements of our approach, Table 10 presents a comparison with the existing ML-based anomaly detection frameworks. Unlike previous approaches, our system integrates XAI

(SHAP) with ML models in an edge-cloud environment, improving both security and real-time detection capabilities.

4.5. Real-world use case - IoT in healthcare

We demonstrate the application of our approach in a real-world use case, specifically in an IoT-based healthcare setting, illustrating how each layer is utilized. We start with the first layer, which covers Healthcare IoT System Architecture and Data Collection. Next, we examine the use of Edge Computing for low-latency anomaly detection in healthcare, followed by ML-based anomaly detection and XAI in the cloud. Finally, we present real-time healthcare monitoring and alerts.

To validate the practical applicability of the proposed ML-based anomaly detection system, its deployment in healthcare IoT (H-IoT) environments was explored. Healthcare is a mission-critical domain in

Table 10

Comparative analysis of anomaly detection methods.

Study	ML model used	Security considerations	Explainability (XAI)	Cloudedge integration	Limitations
(Kansal et al., 2024)	Recurrent Neural Network	Limited	No	Cloud only	Latency issues, Security concerns
(Xiang et al., 2024)	Federated Learning iForest	Yes	No	IoT + Edge	Data privacy, computational cost
(Ismaeel & Elmedany, 2022)	k-means Clustering	No	No	Cloud only	High false positives
(Li et al., 2021)	iForest	Limited	No	Cloud + Edge	Struggles with irregular data
(Mahajan et al., 2021)	XGBoost	Yes	Yes (SHAP)	Cloud	No real-time processing
Proposed Study	iForest + LOF	Multi-layered security	Yes (SHAP for interpretability)	Edge + Cloud optimized	Low latency, high accuracy

which anomaly detection plays a vital role in real-time patient monitoring, security threat detection, and automated clinical decision support.

4.5.1. IoT layer: healthcare IoT system architecture and data collection

IoT-enabled healthcare systems combine wearable biosensors, RPM devices, and ML algorithms with IoT to monitor crucial physiological parameters in real-time. These platforms offer uninterrupted patient surveillance across diverse settings, including hospitals, ICUs, and home environments. The sensor data collected undergoes comprehensive analysis to identify abnormalities in cardiac function, breathing patterns, and variations in blood pressure or glucose levels, facilitating prompt medical intervention to avoid complications.

The infrastructure incorporates various biomedical sensors, such as ECG monitors, pulse oximeters, glucometers, and smart insulin pumps, which constantly transmit physiological information to an EC layer. This layer conducts initial anomaly detection before sending critical alerts to cloud-based ML models for in-depth examination. For example, a wearable ECG device persistently monitors heart rate and electrical activity, recognizing arrhythmias that could signal an imminent stroke or heart failure. Likewise, a CGM tracks blood sugar levels and notifies diabetic patients and their healthcare providers of unusual fluctuations, enabling timely medical action.

To guarantee secure and efficient data transmission, the system uses the MQTT protocol with TLS/SSL encryption, safeguarding sensitive health information from unauthorized access. The cloud platform receives only anonymized and encrypted sensor data, preserving patient privacy while enabling AI-driven predictive analytics for healthcare. This secure and intelligent integration of IoT in healthcare improves proactive patient care, minimizing risks, and improving overall health outcomes.

4.5.2. Edge layer: edge computing for low-Latency anomaly detection in healthcare

In the medical field, where time is of the essence, the rapid identification of health anomalies can be crucial for timely intervention and prevention of serious complications. Conventional healthcare systems that exclusively rely on the Internet of Things often face network delays, which results in slower anomaly detection. To address this issue, EC is now being incorporated into medical monitoring devices, enabling on-site processing of sensor data in real-time. This approach facilitates immediate decision making without dependence on distant servers.

The edge layer functions as the primary anomaly detection mechanism, substantially decreasing the time required to recognize and address critical health issues. For example, if a patient's SpO2 level falls below 88 %, the edge node immediately signals an alert, allowing ICU personnel to provide emergency oxygen therapy without waiting for cloud-based evaluations. This quick response system is enhanced by predefined rule-based thresholds that identify specific medical emergencies. These include a heart rate above 180 BPM, which can indicate tachycardia or cardiac arrest; blood glucose levels exceeding 300 mg/dL, suggesting a risk of hyperglycemic coma in diabetic individuals; and oxygen saturation below 85 %, which can indicate hypoxia and respiratory failure.

In addition to anomaly detection, Edge AI bolsters data protection by utilizing AES-256 encryption, safeguarding sensitive biomedical information during transmission. Moreover, the edge layer performs initial

data filtering, sending only crucial anomalies to the cloud. This selective method optimizes network bandwidth usage and reduces the computational burden on cloud infrastructure, enhancing overall system efficiency. By integrating EC into healthcare systems, medical facilities can improve real-time decision making, improve patient outcomes, and maintain robust data security.

4.5.3. Cloud layer: ML-based anomaly detection and XAI in the cloud

Cloud-based analysis of processed medical data utilizes ML models to identify complex patterns in patient health trends. This ML-driven approach, unlike conventional rule-based medical alarms prone to false positives and healthcare professional alarm fatigue, adjusts to normal patient behavior, enhancing anomaly detection accuracy. The system employs multiple ML models to boost predictive precision.

The iForest algorithm recognizes novel health anomalies not explicitly programmed, such as early sepsis indicators based on fluctuations in body temperature and heart rate. The LOF algorithm identifies localized deviations of vital signs of the patient, such as sudden drops in blood pressure during operations. A Threshold-Based Detection system is also implemented for immediate critical medical emergency alerts, such as spikes in glucose levels in diabetic individuals. For example, if an elderly patient's ECG waveform shows irregular patterns, the iForest algorithm might flag this as a potential atrial fibrillation precursor, allowing physicians to make proactive medication adjustments before a severe cardiac event occurs.

To build trust in AI-driven medical decisions, the system implements XAI through SHAP, ensuring transparent anomaly predictions. SHAP provides insights into anomalous data classifications by offering a breakdown of contributing factors. In the case of a detected cardiac abnormality, SHAP values might indicate that ECG waveform variability contributed 45.2 % to the abnormal score, heart rate spikes 38.6 %, and systolic blood pressure 24.1 %. This transparency enables cardiologists to verify AI-generated alerts before taking action, reducing unnecessary interventions and preventing overmedication. By combining ML-based anomaly detection with explainability, the system improves both patient safety and clinical decision-making (Yazici et al., 2023).

4.5.4. SSPA: real-time healthcare monitoring and alerts

The SSPA serves as an interactive monitoring platform for healthcare providers and distant caregivers to track patient health metrics and receive instantaneous alerts for abnormalities. Shows real-time visualization of vital signs of the patient, such as ECG, blood pressure, oxygen saturation, and glucose levels, on a responsive graph. When life-threatening situations are identified, the system sends alerts via SMS, Email, and in-app notifications. The SSPA employs AI-powered SHAP explanations to help physicians comprehend anomalies prior to making medical judgments. In addition, it provides access to past data to analyze trends and forecast long-term disease progression. As an example, if a patient recovering from surgery experiences an abrupt decrease in blood pressure, the SSPA immediately informs hospital personnel, allowing rapid medical action.

5. Challenges and limitations

Several challenges were encountered during the experiments. The heterogeneity of IoT data sources presents difficulties in pre-processing

and standardization, necessitating rigorous data-cleaning procedures. Moreover, balancing real-time anomaly detection with computational efficiency is a significant challenge given the resource constraints associated with ML processing in CC environments. Furthermore, ensuring secure data transmission and storage requires robust encryption and authentication mechanisms to mitigate potential vulnerabilities.

Despite these promising results, certain limitations of this study merit consideration. System performance may vary in real-world deployments featuring more diverse IoT devices and dynamic data distributions. Furthermore, although this study focused on specific ML algorithms, alternative models may yield different performance outcomes. Future studies should explore alternative CC service models such as Platform as a Service (PaaS) and Software as a Service (SaaS) to assess their impact on system efficiency.

6. Conclusions and future work

This research tackles the issue of improving ML (ML)-based anomaly detection in data security management by combining IoT, CC, and EC. A significant contribution is the creation of an innovative multi-tier architectural model that incorporates various technologies and security protocols. This holistic approach enables efficient data transfer, real-time anomaly identification, and secure communication between IoT devices and cloud services, thus reducing security risks and enhancing system resilience.

The study's primary outcomes showcase the effectiveness of the proposed method in boosting privacy protection, enhancing anomaly detection precision, and ensuring network security through advanced encryption protocols and security certificates. Among the anomaly detection techniques tested, the iForest algorithm proved to be most effective, achieving the highest AUROC score and demonstrating a superior ability to differentiate between normal and anomalous data points. Incorporating diverse security protocols further strengthens network reliability and resistance to potential cyber threats.

These findings have significant implications for IoT, CC, and EC, offering a scalable and adaptable framework suitable for implementation in various application domains, such as industrial automation, healthcare and smart cities. The proposed multi-tier architecture exhibits key features like adaptability, maintainability, and scalability, ensuring robust data management, secure communication channels, and optimized system performance. Consequently, this study highlights a viable approach to enhance ML-based anomaly detection in IoT-cloud interactions.

This research emphasizes the transformative potential of integrating ML, IoT, CC, and EC for anomaly detection and data security. Although substantial progress has been made, further investigation is needed to optimize the proposed framework. Future studies should explore alternative ML algorithms, ensemble learning techniques, and deep learning models to improve detection accuracy and system robustness. Moreover, expanding this research to include other cloud service models, such as PaaS and SaaS, would provide deeper insights into their impact on real-time anomaly detection and resource management.

Furthermore, incorporating FC and edge intelligence into the existing framework could further decrease latency and improve real-time decision making. These advances would be particularly beneficial for mission-critical applications where timely anomaly detection is crucial to operational reliability and security.

While the current study focuses on hardware-based IoT sensors for anomaly detection, it does not incorporate video-based surveillance due to the non-invasive nature of our research and constraints. However, the significant potential of video and image-based anomaly detection techniques is acknowledged, including computer vision-based models for security monitoring and behavior analysis. In the next phase of research, the integration of video-based anomaly detection methods is planned to further enhance the robustness of the framework, particularly for

security-sensitive environments such as healthcare and industrial monitoring.

As IoT-generated data continues to grow in complexity and volume, researchers must explore more advanced integration strategies to enhance data security, privacy, and computational efficiency. Given the increasing reliance on cloud-based infrastructure, future research should prioritize resource allocation optimization, adaptive security mechanisms, and anomaly detection based on federated learning to address emerging challenges in cloud environments. The continued focus on technological innovation and security-driven improvements will contribute to the development of more efficient, scalable, and secure data management systems, strengthening the resilience of modern IoT ecosystems against evolving cyber threats.

CRedit authorship contribution statement

Sultan Baimukhanov: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Visualization; **Hashim Ali:** Conceptualization, Methodology, Validation, Writing – review & editing; **Adnan Yazici:** Supervision, Formal analysis, Writing – review & editing.

Data availability

The authors do not have permission to share data.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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